

RAAHAT

Regime-Aware Adjustment of Heavy-rainfall Alerts & Thresholds

Team briefing — everything you need to build the deck. Written for non-technical readers. Every technical word is explained the first time it appears. Slide-by-slide guidance starts on page 4.

What the project is, in one paragraph. When heavy rain is coming, someone in a government office must decide whether to warn a district. They use computer weather predictions, and those predictions are wrong in ways that repeat. RAAHAT is a layer that sits on top of those predictions and corrects them before a warning goes out. It does not forecast weather. It cleans up somebody else's forecast.

The one idea that makes it different. Weather models don't make random mistakes — the mistake depends on what kind of weather is happening. The same model over-predicts drizzle in a dry spell and badly under-predicts a Bay of Bengal storm. Everyone today corrects them the same way regardless. We work out what kind of weather is coming first, then apply the correction that situation deserves.

HEADLINE NUMBERS (all measured on real data the model had never seen)

Measure	Raw European forecast	RAAHAT	Change
Heavy-rain detection score	0.054	0.180	3.3x better
Catches heavy rain (of 168 events)	1 in 15	1 in 3	POD 0.065 to 0.339
Rainfall error (mm/day)	13.76	12.64	8% lower
Agreement with reality	0.502	0.578	higher
False alarms	0.756	0.723	slightly fewer

Tested on August–September 2025, 197 districts, 12,017 district-days. The test data was locked away and opened exactly once, after the model was frozen — so these numbers cannot have been tuned to look good.

THE RESULT WE LEAD WITH

Our clever step — recognising the weather situation — helps **most where the theory says it should**, and not elsewhere. That pattern is much stronger evidence than a flat improvement everywhere.

Weather situation	Raw forecast	Without our step	With our step
Low-pressure system (storm)	0.023	0.171	0.232 (+36%)
Ordinary unsettled weather	0.046	0.139	0.142 (+2%)

In storm conditions the correction is **ten times** better than the raw forecast. In ordinary weather our extra step adds nothing — which is exactly right, because there is no distinctive situation to recognise.

Words your audience may not know

Use these definitions on slides or in the script. Judges will ask.

Term	What it means
Post-processing	Correcting someone else's forecast after they produced it. This is what we do. We are not a weather model.
NWP (Numerical Weather Prediction)	The big physics simulations that produce forecasts — run by Europe, Germany, USA, India. We consume their output.
Lead time / Day-3 forecast	How far ahead. 'Day 3' means predicted three days before it happens. Warnings are usually decided around day 3.
Regime	The kind of weather situation. We use four: active monsoon spell, break (a lull), a low-pressure system (storm), and ordinary unsettled.
LPS / Low-pressure system	A monsoon depression — the spinning storms that drift in from the Bay of Bengal and cause most severe flooding.
POD (Probability of Detection)	Of all the heavy-rain events that happened, what fraction did we warn about? Higher is better.
FAR (False Alarm Ratio)	Of all our warnings, what fraction were wrong? Lower is better.
CSI (Critical Success Index)	A single score combining both — rewards catching events, penalises false alarms. Our headline number.
RMSE	Average size of the error in millimetres. Lower is better.
Calibration	Does '70% chance' actually happen 70% of the time? A warning system that says 70% and means 30% is dangerous.
Held-out test	Data deliberately hidden from the model during building, used once at the end. The only honest measure of how good something is.
Ablation	Switching one part off to see whether it mattered. We switched off our regime step to prove it earns its place.

How the system works, in five steps

- **1. Read someone else's forecast.** We take the European (ECMWF) and German (ICON) weather models' predictions for the next 1–5 days.
- **2. Work out what kind of weather is coming.** From the predicted pressure patterns and rainfall across a grid over India, we identify whether a storm, an active spell, a lull, or ordinary weather is due.
- **3. Apply the matching correction.** Each situation gets the adjustment it deserves, learned from two monsoons of past mistakes.
- **4. Produce a range, not a single number.** Instead of '180 mm' we give 'most likely 180 mm, with a 41% chance of exceeding 204.5 mm'. Those cut-offs are IMD's own official heavy / very heavy / extremely heavy boundaries.
- **5. Turn it into a colour.** Green / yellow / orange / red, using a dial the user controls for how cautious they want to be.

Two safety rules worth mentioning on stage

- **It can never quietly lower a warning.** If the raw forecast would have said orange, we cannot say yellow — unless we are genuinely confident it is dry. A tool that silently downgrades a real warning is worse than no tool.
- **Warnings don't flicker.** A district cannot flip orange–yellow–orange between updates. A real escalation passes through untouched.

What we must say honestly (this wins marks, it doesn't lose them)

Judges trust teams that state their own limits before being asked. Every item here is measured, not guessed.

1. We miss most heavy-rain events

Of 168 heavy-rain events, **108 (64%) were never flagged** — the model correctly knows about 1% of days will be heavy, but often cannot say *which* 1%. We improved detection from 1-in-15 to 1-in-3; that is real progress, not a solved problem. Say the 1-in-3, not 'we detect heavy rain'.

2. The clever part only clearly helps in storms

Averaged across all weather, our regime step adds very little and is within measurement noise. Split by situation, it adds 36% in low-pressure-system conditions. **Quote the split figure, and say it is a split figure.**

3. The probabilities are under-confident

When the system said '29% chance of heavy rain', it actually happened 63% of the time. We diagnosed why — the calibration step had too few examples and flattened the top of its range — and fixed it (error down 4.6x). The fix could not be re-tested on the same held-out data, so the reported numbers are the conservative ones.

4. Two of six weather situations cannot be detected

Western Disturbances and coastal-easterly regimes need upper-atmosphere data that is not freely available. They are shown as zero rather than quietly dropped.

5. Small sample

Two monsoons, 197 districts. For the rarest rainfall there simply are not enough events, and we mark those cells 'insufficient data' rather than printing a confident-looking number built on four events.

The framing that protects us

We are not competing with IMD. We consume the same kind of output they do and calibrate it. Their warnings mean 'heavy rain at isolated places within a district'; ours estimates a district-wide average. Different quantities. Say this on the slide before a judge says it to you — it turns a weakness into evidence that we understand the domain.

Where the data comes from (all free, no payment, no special access)

Data	Source	Notes
Actual rainfall (the truth)	India Meteorological Department, Pune	Official 0.25° gauge-based grid, 16 years (2010–2025)
Forecasts	ECMWF (Europe) + ICON (Germany) via Open-Meteo	Free, no key, CC BY 4.0 licence
Pressure patterns	ERA5 reanalysis via Open-Meteo	Used to spot storms
District boundaries	Datameet, Census 2011	Open licence (ODbL)

Required credit line for the deck and any screen: “Weather data by Open-Meteo.com (CC BY 4.0). Observed rainfall: India Meteorological Department, Pune.”

What is built, and what comes next

Use this to decide what the deck may claim. Anything in the second or third table must be described as planned, not shown as working.

BUILT AND MEASURED — safe to claim outright

Piece	What it does	Evidence
Data pipeline	IMD actual rainfall + European and German forecasts, 197 districts, two monsoons	0% missing; matched a known flood event
Weather-situation recogniser	Identifies storm / active spell / lull / ordinary, 1–5 days ahead	Finds real storms, e.g. the Aug 2024 Gujarat depression
The correction model	Adjusts the forecast per situation; outputs a range plus probabilities	3.3x detection vs raw forecast
Comparison tests	Five alternatives tested side by side	Table on Slide 4
Held-out evaluation	Tested once on hidden data, model frozen first	Full audit trail recorded
Warning colours	Green/yellow/orange/red, never silently downgrades	16 automated tests
Explanations	Top three reasons for each forecast, plus similar past days	12 automated tests
Error Atlas	Measures how wrong forecasts are per situation and region	Built; needs more data to fill every cell

NEXT — planned before the finale, describe as 'in progress'

Piece	Why it matters	State
The screen	Map, district panel, sliders — what judges actually look at	Project set up; no code written yet
All-India expansion	197 to 641 districts, ~2.8x more events; would fill the Atlas and the empty rows of our evidence table	Started; data fetch stalled at 12%
Offline demo mode	So the presentation cannot fail on venue wifi	Not started
Web service check	Confirm the service serves the screen correctly	Written, not yet verified
2023 as extra training data	A third monsoon, single-lead only	Confirmed available; not ingested

LATER — mention as future work only

Piece	Why it is not done
A more advanced correction model (CSGD mixture)	The current one already works; this would refine the rainfall range. We found and documented a technical obstacle and two ways round it.
Benchmark against IMD's own issued warnings	Needs a registration on IMD's public API that has not been completed. This is the single most time-critical item — the archive cannot be backfilled, so every day without it is lost permanently.
Hindi bulletins, mobile view, more regions	Straightforward but not yet needed for the demo.

***How to present the roadmap.** Put the built/next split on Slide 4 (Feasibility) as a small two-column box. A team that shows a credible next step scores better than one that implies everything is finished — and far better than one caught claiming something that does not run.*

Slide-by-slide guidance

One idea per slide. Under 25 words of body text per slide. The four IMD warning colours (green/yellow/orange/red) should be the only strong colours anywhere in the deck.

SLIDE 1 — INTRODUCTION / PROBLEM

The one message: Weather models make the same mistake every time a particular weather pattern occurs — so the mistake is predictable, and therefore fixable.

Put on the slide

- A single hook number: “**At three days out, the operational forecast catches one heavy-rain event in fifteen.**”
- One sentence on what we build: a correction layer between the forecast and the district warning desk.
- One line stating we do NOT build a weather model and do NOT compete with IMD.
- Who uses it: IMD duty forecaster, District Collector / disaster management officer.

***Best visual:** one scatter plot — predicted rainfall against what actually fell, with dots coloured by weather situation, so the different situations visibly sit on different lines. That single picture is the whole argument.*

Do NOT put: team photos, college logos, a generic 'India faces floods' paragraph, or more than three lines of text.

SLIDE 2 — TECHNICAL STACK

The one message: Everything is free, open and runs on an ordinary laptop — no GPU, no paid data, no special government access.

Put on the slide

Layer	What we use	Status
Data	IMD gridded rainfall; ECMWF + ICON forecasts via Open-Meteo	Working — free, open licences
Learning	LightGBM (decision-tree model), scikit-learn	Working — trains in minutes, CPU only
Storage	Parquet files + DuckDB	Working — no database server
Service	Python FastAPI	Written , end-to-end check pending
Screen	React + MapLibre map	Planned — set up, not yet built
Checks	74 automated tests	Working — all passing

Keep the status column on the slide. Judges are far harder on a team that implies something works than on one that says plainly what is built and what is next. See the build-status page for the full picture.

***Say out loud:** “No GPU anywhere. Trains in minutes. Every input is free and openly licensed.” That is a genuine competitive claim — many teams cannot say it.*

Explain if asked why not deep learning: we have two monsoons of data. A large neural network would simply memorise it. Decision trees are the right size of tool for the size of data we have — and we can explain every prediction, which a neural network cannot.

SLIDE 3 — ARCHITECTURE / HOW IT WORKS

The one message: We predict the weather situation from the forecast itself — so it works three days ahead, not just in hindsight.

Draw four boxes left to right. Not ten.

1. DATA	2. RECOGNISE	3. CORRECT	4. DECIDE
Forecasts from ECMWF & ICON. Actual rainfall from IMD. Pressure patterns.	Work out the weather situation for each day, 1–5 days ahead, from the forecast — not from what already happened.	Apply the correction that situation deserves. Output a range of possible rainfall, not one number.	Convert to a probability of crossing IMD's official thresholds, then to a warning colour.

Three callouts to annotate the diagram with — these are what a judge should read even if they read nothing else:

- On box 2: *“The weather situation is FORECAST, not diagnosed after the fact. That is what makes it usable at day 3.”*
- On box 3: *“We blend situations rather than picking one. Real days are mixtures — a storm inside an active spell is one day, not two.”*
- On box 4: *“The caution level is a dial the operator controls, not a number we hard-coded.”*

Good second visual: a bar chart of the weather-situation probabilities changing as a storm approaches, with the correction size tracking underneath. That is the most watchable fifteen seconds you have.

SLIDE 4 — FEASIBILITY AND VIABILITY

The one message: It is already built and already measured on data it had never seen. This is not a proposal.

Put the evidence table on the slide — this is the most important table in the deck:

What was tested	Detection score	Verdict
Long-term averages (do nothing)	0.000	no skill
Raw European forecast	0.054	the starting point
Blending several models (what's done today)	0.069	barely helps
Standard textbook correction	0.138	helps, but worsens accuracy
RAAHAT without our clever step	0.171	post-processing works
RAAHAT, full	0.180	3.3x the raw forecast

- **Feasible:** free data, no GPU, trains in minutes, 74 automated tests.
- **Honest:** the test data was locked and opened once, after the model was frozen. Point at the row where we did not improve and say so.
- **Scalable:** runs as a nightly batch job. Rainfall forecasts update four times a day — nothing here needs to be real-time.
- **Risks we name first:** two monsoons of data; we miss 64% of events; two weather situations we cannot yet detect.

Add a small two-column roadmap box to this slide — it is the natural home for it and it answers 'what would you do next?' before a judge asks:

Working today	Next
Data pipeline, situation recogniser, correction model, warning colours, explanations, held-out evaluation	The forecaster's screen; all-India (197 to 641 districts); offline demo mode; benchmark against IMD's own issued warnings

SLIDE 5 — IMPACT AND BENEFITS

The one message: Better warnings, earlier, with fewer false alarms — using forecasts India already produces.

Put on the slide

Benefit	The evidence
Catches 5x more heavy-rain events	1 in 15 to 1 in 3 at three days ahead
Buys about three days of lead time	Our day-5 forecast is as accurate as the raw day-2 forecast
Biggest gain exactly where it matters	+36% in low-pressure-system conditions — the storms that cause most flood deaths
Fewer unnecessary red alerts	False alarms fall from 0.756 to 0.514 in the conservative setting
Costs almost nothing to run	Free data, ordinary laptop, minutes to train

Who benefits

- **Duty forecasters** — a district table with corrected rainfall, probabilities, a suggested colour, and the reason behind it.
- **District Collectors and disaster management officers** — a map and a plain-language bulletin with a confidence number.
- **People in affected districts** — fewer missed warnings, and fewer false red alerts that erode trust in the next real one.
- **Rain-fed farmers** — better three-day guidance for sowing, harvesting and irrigation decisions.

The closing line: “Every input is free and open. It trains in minutes on a laptop with no GPU. It sits upstream of India’s forecasting systems, not against them.”

Final reminders for whoever builds the deck

- **One idea per slide.** Under 25 words of body text.
- **Every number carries its sample size.** Write “0.180 (168 events)”, not “0.180”.
- **Use only the four IMD warning colours** as strong colours. Everything else grey. When a district is red it must be the only red thing on the screen.
- **Show one place we did not improve.** It buys more credibility than anything else in the presentation.
- **Never imply live real-time capability.** Everything shown is replay of real past dates, and saying so is a strength.

Generated from the project’s own results files. All figures traceable to results/final/ and models/frozen/. Prepared for the RAAHAT team, SIH.